Ensemble Learning - Theory Answers (Q1-Q5, Q10)

Q1. What is Ensemble Learning?

Ensemble Learning is a machine learning technique that combines multiple models to produce a stronger and more accurate predictive model. The key idea is that a group of models often performs better than a single model.

Q2. Difference between Bagging and Boosting

Bagging trains models independently on different bootstrap samples and combines their predictions. Boosting trains models sequentially, where each new model focuses on correcting previous errors.

Q3. What is Bootstrap Sampling?

Bootstrap sampling is a method of randomly selecting observations from a dataset with replacement. It is used in Bagging and Random Forest to create diverse training subsets.

Q4. What are Out-of-Bag Samples?

Out-of-Bag samples are data points not selected in a bootstrap sample. OOB score provides an internal estimate of model performance without needing a separate validation set.

Q5. Decision Tree vs Random Forest Feature Importance

A Decision Tree calculates importance from a single tree and may be unstable. A Random Forest averages importance across many trees, making it more reliable and robust.

Q10. Loan Default Prediction Using Ensemble Learning

Choose Bagging when reducing variance and overfitting is the goal; choose Boosting when maximizing predictive accuracy. Handle overfitting through hyperparameter tuning, cross-validation, pruning, and regularization. Select Decision Trees as base learners because they capture nonlinear patterns. Evaluate models using k-fold cross-validation and metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC. Ensemble methods improve decision-making by increasing predictive accuracy, reducing risk, and enabling better loan approval decisions.